

last time: norms, inner products, orthonormal bases, Gram-Schmidt

today: orthogonal complement, example using Gram-Schmidt
linear operators, projections, closest point property

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Suppose V_0 is a subspace of V (actually, V_0 can be any subset of V)

def: The orthogonal complement $V_0^\perp$ of V_0 is

$$V_0^\perp = \{u \in V : \langle u, v \rangle = 0 \quad \forall v \in V_0\}$$

why is this a subspace?

$$x, y \in V_0^\perp, \alpha \in K$$

$$\Rightarrow \langle \alpha x + y, v \rangle$$

$$= \alpha \langle x, v \rangle + \langle y, v \rangle = 0 \quad \forall v \in V_0$$

$$\Rightarrow \alpha x + y \in V_0^\perp$$

so $V_0^\perp$ is closed under linear combinations

Example: find an orthonormal basis for the orthogonal complement of $V_0 = \text{span} \left\{ \begin{pmatrix} 3 \\ -6 \\ 1 \end{pmatrix} \right\}$ in $\mathbb{C}^3$

solution: $V_0^\perp = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : \left\langle \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \begin{pmatrix} 3 \\ -6 \\ 1 \end{pmatrix} \right\rangle = 0 \right\}$

$$3x - 6y + z = 0$$

step 1: find any basis for $V_0^\perp$: $\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}$

step 2: do Gram-Schmidt

$$e_1 = \frac{\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}}{\sqrt{4+1+0}} = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

$$E_2 = \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} - \left\langle \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, e_1 \right\rangle e_1$$

$$= \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} - \frac{-2}{\sqrt{5}} \cdot \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} + \begin{pmatrix} 4/5 \\ 2/5 \\ 0 \end{pmatrix} = \begin{pmatrix} -1/5 \\ 2/5 \\ 3 \end{pmatrix}$$

$$e_2 = \frac{E_2}{\|E_2\|} = \frac{5E_2}{\|5E_2\|} = \frac{1}{\sqrt{1+4+225}} \begin{pmatrix} -1 \\ 2 \\ 15 \end{pmatrix}$$

$$= \frac{1}{\sqrt{230}} \begin{pmatrix} -1 \\ 2 \\ 15 \end{pmatrix}$$

Linear operators Let V, W be vector spaces over $K = (\mathbb{R} \text{ or } \mathbb{C})$

$$T: V \rightarrow W \text{ is linear if } \begin{aligned} T(u+v) &= Tu + Tv & \forall u, v \in V \\ T(\alpha v) &= \alpha Tu & \forall u \in V, \alpha \in K \end{aligned}$$

Tu means $T(u)$ (it's a function)

examples: (1) $A: \mathbb{C}^n \rightarrow \mathbb{C}^m$ ($m \times n$ matrix)

$$(Ax)_i = \sum_{j=1}^n A_{ij} x_j \quad (1 \leq i \leq m)$$

$\uparrow$
matrix entries

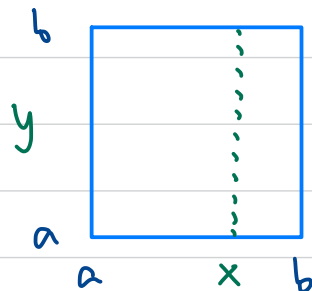
matrix/vector sizes must match:

$$m < n: \begin{pmatrix} A \end{pmatrix} \begin{pmatrix} x \end{pmatrix} = \begin{pmatrix} b \end{pmatrix}, \quad m > n: \begin{pmatrix} A \end{pmatrix} \begin{pmatrix} x \end{pmatrix} = \begin{pmatrix} b \end{pmatrix}$$

$$(2) \quad \mathcal{I}: L^2(a, b) \rightarrow L^2(a, b)$$

$$\mathcal{I}f(x) = \int_a^b \underbrace{K(x, y)}_{\text{the kernel of the integral operator}} f(y) dy$$

the kernel of
the integral operator



kernel has two meanings

1. the function $G(x,y)$ that you integrate against
2. the set of vectors that are mapped to zero (i.e., the nullspace of G)

example: the solution of the boundary value problem

$$\begin{cases} -u''(x) = f(x) \\ u(0) = u(1) = 0 \end{cases}$$

is given by

$$u(x) = \int_0^1 \underbrace{G(x,y)}_{\text{Green's function}} f(y) dy$$

$$G(x,y) = \begin{cases} x(1-y) & 0 < x < y < 1 \\ y(1-x) & 0 < y < x < 1 \end{cases}$$


e.g. if $f(x) \equiv 1$ then

$$u(x) = \int_0^x y(1-x) dy + \int_x^1 x(1-y) dy = \frac{1}{2}(x - x^2)$$

$$\begin{aligned} -u'' &= 1 \checkmark \\ u(0) &= u(1) = 0 \checkmark \end{aligned}$$

$$\frac{x^2}{2}(1-x) \quad x(y - \frac{y^2}{2}) \Big|_x^1 = \frac{x}{2} - x^2 + \frac{x^2}{2}$$

example: convolution

$$g * f(x) = \int_{-\infty}^{\infty} g(x-y) f(y) dy$$

(more later)

special case, $K(x,y) = g(x-y)$

$$(3) \quad T: \mathbb{C}^N \rightarrow V$$

$$Tc = \sum_{n=1}^N c_n \varphi_n$$

$\varphi_1, \dots, \varphi_N$ lin. indep. vectors in V

The range of T is the subspace of V spanned by $\{\varphi_n\}_{n=1}^N$

example: $V = L^2(0,1)$, $N=4$

$$(\varphi_n)_{n=0}^3 = \{1, x, x^2, x^3\}, \quad c \in \mathbb{C}^4$$

$$Tc = \sum_{n=0}^3 c_n x^n$$

(4) linear functionals: $\ell: V \rightarrow \mathbb{C}$ (target vector space is $\mathbb{C}$ or $\mathbb{R}$)

examples (a) $\ell(f) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$ (mean value)

The objects in V are often functions. A "functional"

is a scalar-valued "function of functions"

(b) If $v \in V$, $\ell_v(u) = \langle u, v \rangle$

(or $\ell_v = \langle \cdot, v \rangle$)
"dot notation"
The dot is waiting for
a "u" to act on

(5) Orthogonal projection: If $e_1, \dots, e_N$ is an orthonormal basis
for a subspace V_0 of V , we define $P: V \rightarrow V_0$ by

$$Pu = \sum_{n=1}^N \langle u, e_n \rangle e_n$$

properties

1. P is linear

$$P(\alpha u + v) = \sum_n \langle \alpha u + v, e_n \rangle e_n = \sum_n [\alpha \langle u, e_n \rangle e_n + \langle v, e_n \rangle e_n] \\ = \alpha P u + P v$$

2. for $u, v \in V$, $\langle Pu, v \rangle = \langle u, Pv \rangle$ (P is self-adjoint)

$$\langle Pu, v \rangle = \left\langle \sum_{n=1}^N \langle u, e_n \rangle e_n, v \right\rangle = \sum_{n=1}^N \langle u, e_n \rangle \langle e_n, v \rangle \\ = \sum_{n=1}^N \overline{\langle v, e_n \rangle} \langle u, e_n \rangle = \left\langle u, \sum_{n=1}^N \langle v, e_n \rangle e_n \right\rangle = \langle u, Pv \rangle$$

3. if $v_0 \in V_0$, $Pv_0 = v_0$

$$\text{let } v_0 = \sum_{k=1}^N \alpha_k e_k. \text{ Then } Pv_0 = \sum_{n=1}^N \left\langle \sum_{k=1}^N \alpha_k e_k, e_n \right\rangle e_n \\ = \sum_{n=1}^N \sum_{k=1}^N \alpha_k \underbrace{\langle e_k, e_n \rangle}_{\delta_{kn}} e_n = \sum_{n=1}^N \alpha_n e_n = v_0$$

4. $u - Pu \perp v_0$ for all $u \in V, v_0 \in V_0$

$$\langle u - Pu, e_k \rangle = \langle u, e_k \rangle - \left\langle \sum_n \langle u, e_n \rangle e_n, e_k \right\rangle \\ = \langle u, e_k \rangle - \sum_n \langle u, e_n \rangle \underbrace{\langle e_n, e_k \rangle}_{\delta_{nk}} \\ = \langle u, e_k \rangle - \langle u, e_k \rangle = 0$$

so if $v_0 \in V_0$ then $\langle u - Pu, v_0 \rangle = \sum_k \alpha_k \underbrace{\langle u - Pu, e_k \rangle}_0 = 0$

Alternative proof (version presented in class)

$$\begin{aligned}\langle u - Pu, v_0 \rangle &= \langle u, v_0 \rangle - \langle Pu, v_0 \rangle \\ &= \langle u, v_0 \rangle - \langle u, Pv_0 \rangle && \text{(by 2)} \\ &= \langle u, \underbrace{v_0 - Pv_0}_0 \rangle = 0 && \text{(by 3)}\end{aligned}$$

geometrical significance: closest point property

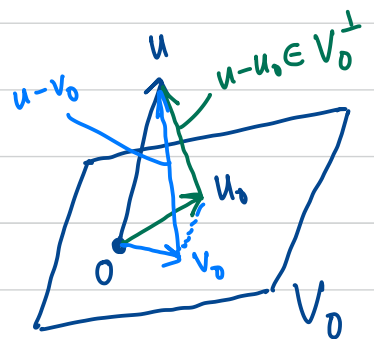
$u_0 = Pu$ is the closest point to u in V_0

Theorem: Let V_0 be a finite-dimensional subspace of V . Then

$$\langle u - u_0, v_0 \rangle = 0 \quad \forall v_0 \in V_0 \quad (\Leftrightarrow) \quad \|u - u_0\| = \min_{v_0 \in V_0} \|u - v_0\|$$

proof: $\Rightarrow$ Suppose $v_0 \in V_0$

$$\text{then} \quad u - v_0 = \underbrace{u - u_0}_{\text{orthogonal to everything in } V_0} + \underbrace{u_0 - v_0}_{\text{in } V_0}$$



$$\therefore \langle u - u_0, u_0 - v_0 \rangle = 0$$

$$\begin{aligned}\text{Pythagorean theorem: } \|u - v_0\|^2 &= \|u - u_0\|^2 + \|u_0 - v_0\|^2 \\ &\geq \|u - u_0\|^2\end{aligned}$$

$$\therefore \|u - v_0\| \geq \|u - u_0\| \quad \text{with equality iff } v_0 = u_0$$

$\Leftarrow$ next time